AUTOMATED DAILY LITERATURE DIGEST • PhD RESEARCH WATCH

Particulate Matter
Monitoring & Health Effects

Issue 2026-10-12 | Entry window 12 October 2026 (full day) | 24 records screened in scope

24	9	8	2	6
RECORDS IN SCOPE (40 EXCLUDED)	OF 10 SUBTOPIC BANDS POPULATED	EFFECT ESTIMATES WITH A CI	TIER-A RECORDS	RECORDS CARRIED ON METADATA ALONE

Signal of the day

346.5 million admissions across six countries: short-term $PM_{2.5}$ raises cancer, mental-disorder and nervous-system admissions more than respiratory ones, with lags out to 21 days.

949 communities in Australia, Brazil, Canada, Chile, New Zealand and Taiwan, 2000–2019, quasi-Poisson time series pooled by meta-analysis. Per 10 $\mu g\,m^{-3}$, all-cause admissions **RR 1.080** (1.074–1.086); malignant neoplasms **1.108** (1.094–1.123), mental disorders **1.100**, nervous system **1.085** — all above respiratory, **1.067** (1.060–1.074). The attributable fraction against the WHO daily guideline fell from 8.9 % (2000–09) to 3.9 % (2010–19). **Where it is weak:** a 21-day distributed lag on cancer admissions is long enough to absorb scheduling and elective-admission structure that has nothing to do with acute exposure, and the pooled RR is reported alongside "substantial heterogeneity" that the abstract does not quantify. The non-respiratory ranking is the result worth testing, not the headline 8 %.

What changed today

- **Composition beats mass, twice.** A 39.7-million-person Medicare preprint puts the wildfire-smoke *carbonaceous* fraction at HR 1.090 (1.077–1.103) per $\mu g\,m^{-3}$ for incident ADRD against 1.029 for smoke $PM_{2.5}$ mass; in Lanzhou, PAHs and Tl/Pb/Mn drive cardiovascular mortality in the traffic district and nitrate/chloride/sulfate in the industrial one. Neither exposure is measurable by an optical node.
- **Sensor-built exposure reaches African epidemiology.** The Kinshasa case-crossover (13,049 admissions) builds its $PM_{2.5}$ series from PurpleAir and QuantAQ nodes fused with bias-corrected GEOS-CF — OR 1.01 (1.00–1.03) per 10 $\mu g\,m^{-3}$, with no reported local validation of the fused series.
- **Two instrumentation results worth keeping.** A fixed distance graph beat learned, wind-informed and hybrid graphs for 12-city $PM_{2.5}$ forecasting in Bangladesh; and LP-DOAS in Fairbanks shows inversion layers only 20–40 m deep persisting for days — a rooftop and a breathing-height node sample different air.
- **A personal-monitor null that should not be read as a null.** Gambian children in solid-fuel homes logged a geometric-mean personal $PM_{2.5}$ of 12.6 $\mu g\,m^{-3}$, and colonisation showed no association; an exposure that low in wood-smoke households points at the monitor, not at the biology.
- **Three held records enter.** The EHP biological-aging cohort, the Nat Commun congenital-heart-defect proteomics study and the alveolar-organoid toxicology paper, held from the 9 October issue when authoring was interrupted, are carried here.

Window continuity

The 11 October issue closed with `last_entry_date` at 11 October. This issue collects **12 October in full**, continuous with it. This run began at 19:46 CDT on 15 October and backfills 12, 13 and 14 October as three separate issues; 15 October is not opened under the 22:00 rule. The 9–11 October issues and the W41 weekly, built on 12 October but never committed, were proofed and pushed at the start of this run.

Corpus analytics

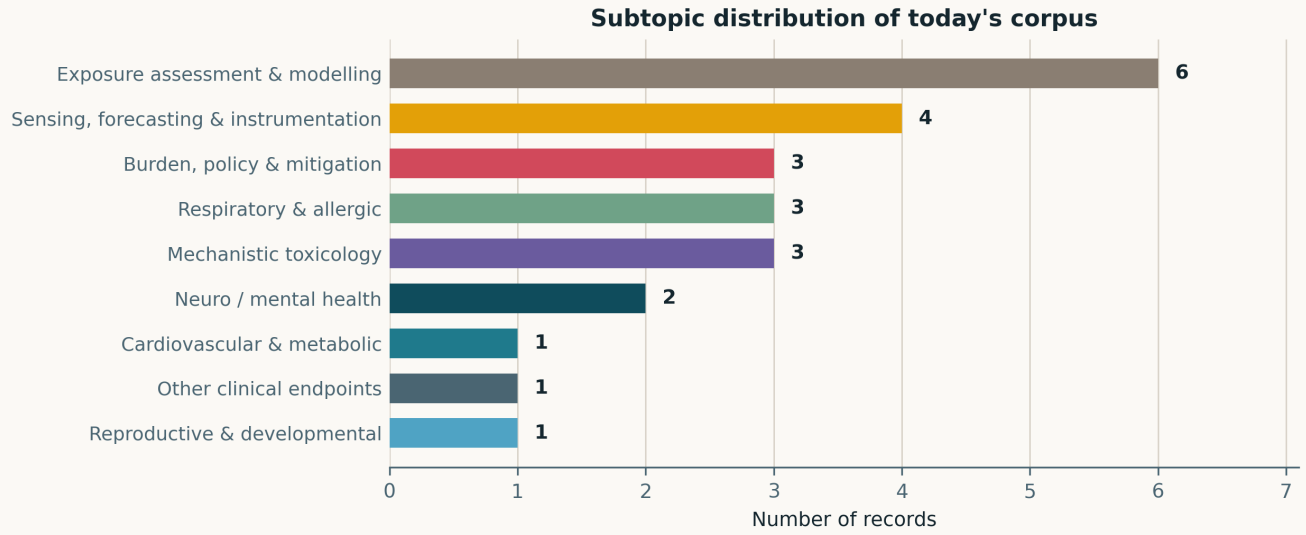


Fig. Subtopic assignment of the 24 in-scope records across nine of ten bands. Exposure assessment leads (6) and sensing is second (4), carried by a graph forecaster, a respirable-dust real-time-vs-gravimetric comparison, a dust-storm PM_{10} forecaster and the Fairbanks inversion profiles.

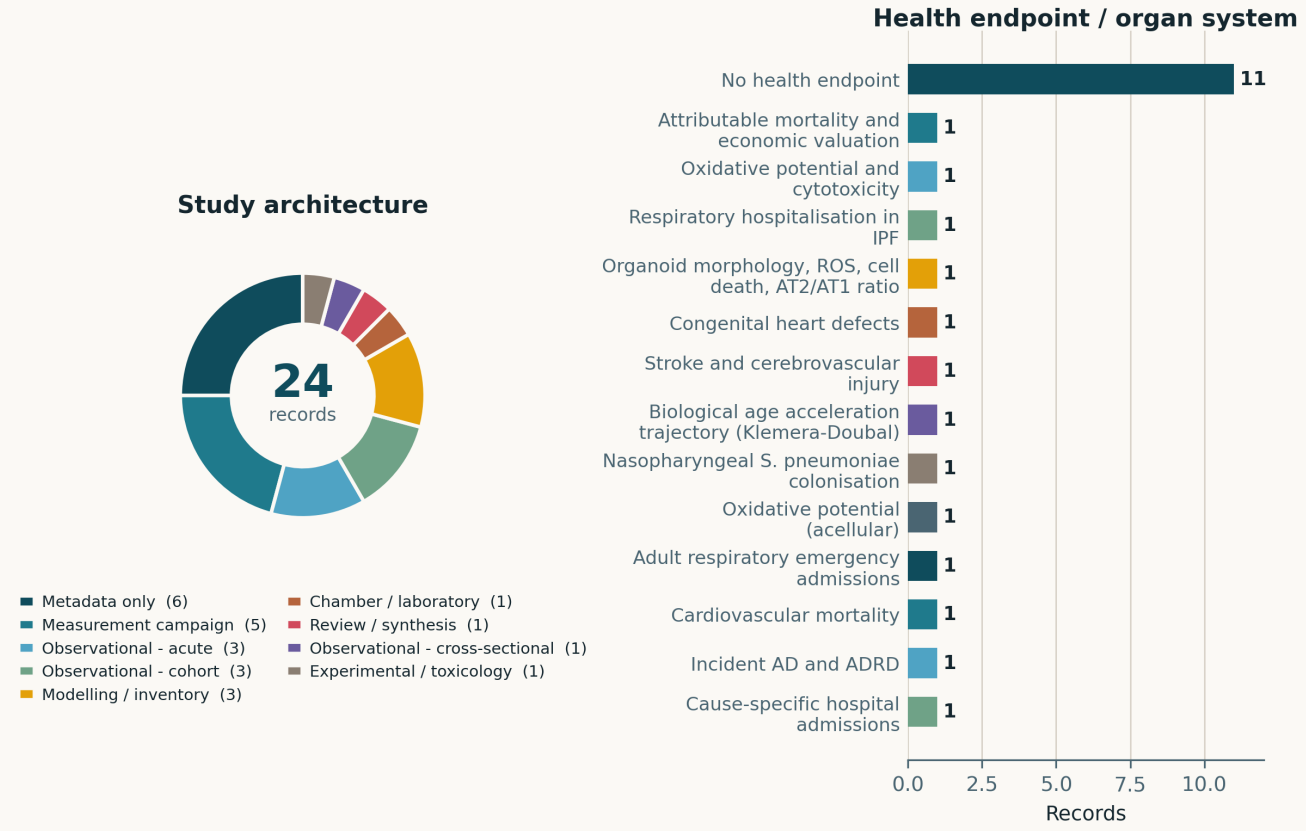


Fig. Left — architecture, including six metadata-only records (25 %), all Elsevier deposits without abstracts at harvest. Right — health endpoints: admissions dominate, with dementia, congenital heart defects and biological aging as single records.

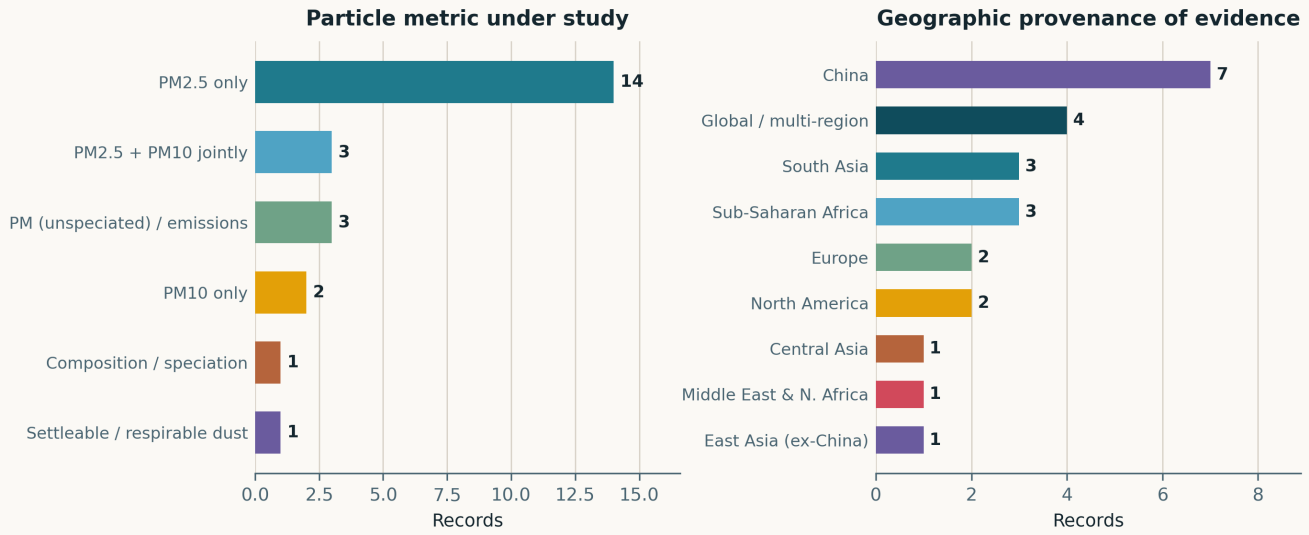


Fig. Left — particle metric: $PM_{2.5}$ dominates; composition-resolved records (smoke carbon, components, metals, WSON) are a quarter of the issue. Right — geography: China leads, with records from the DR Congo, Gambia, Bangladesh, Kazakhstan and Saudi Arabia.

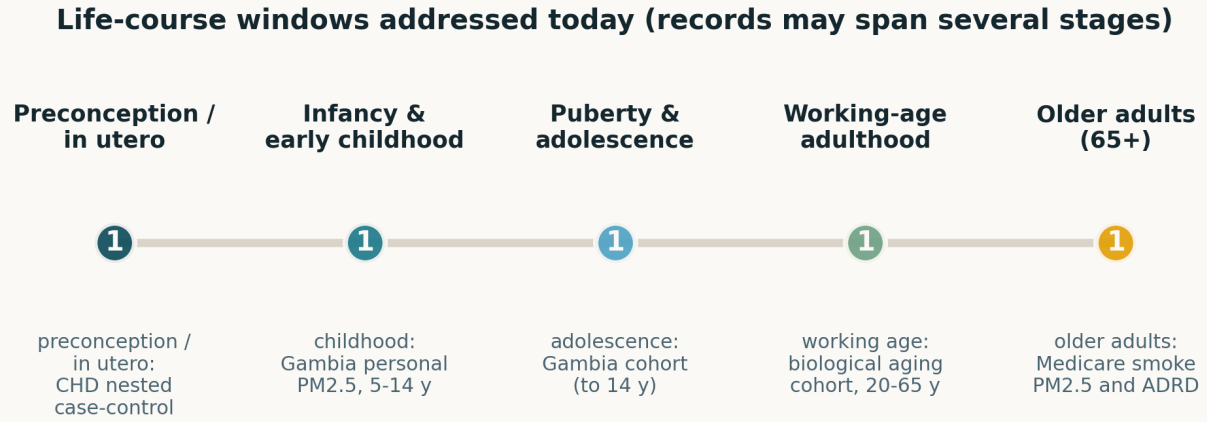


Fig. Life-course windows: all five occupied for the first time since early October — the CHD periconception study, Gambian children 5–14, a 20–65 aging cohort and Medicare beneficiaries.

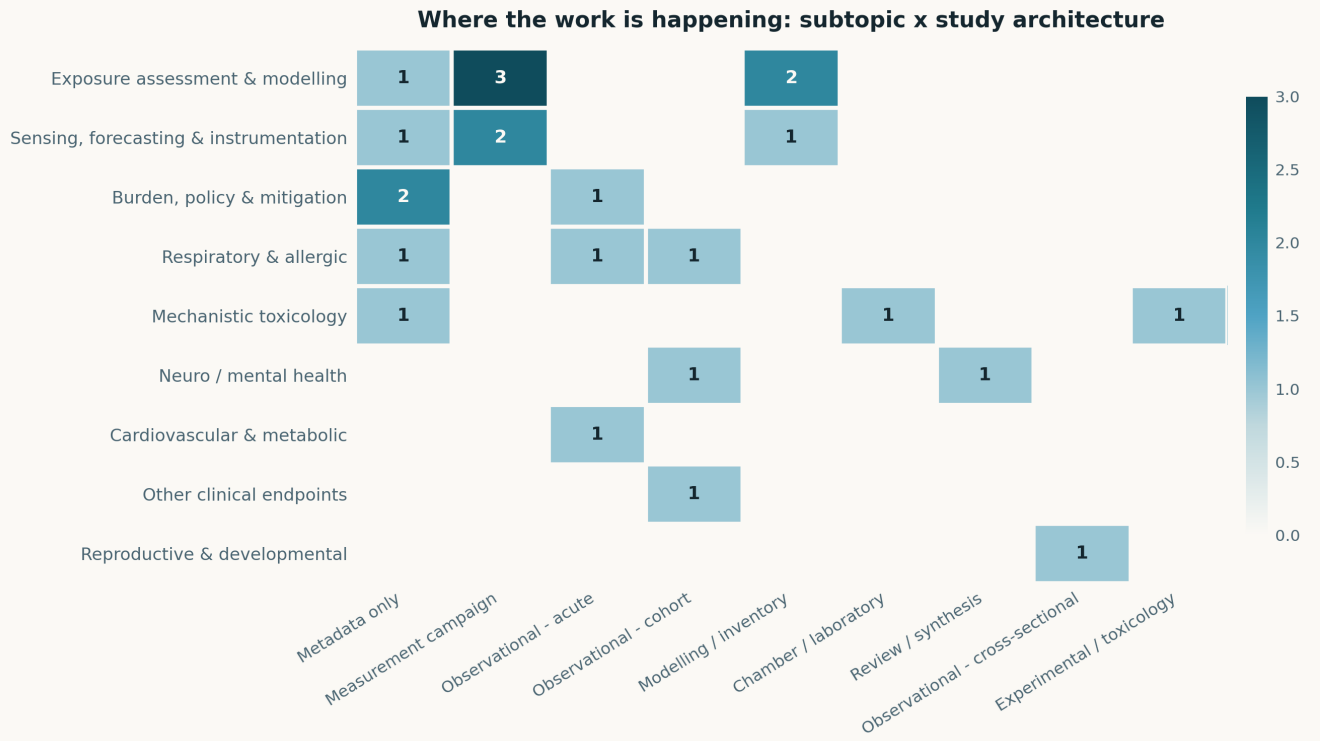


Fig. Subtopic × study architecture. Sensing records are machine-learning or field measurement; the health side is spread across time series, cohorts and one nested case-control.

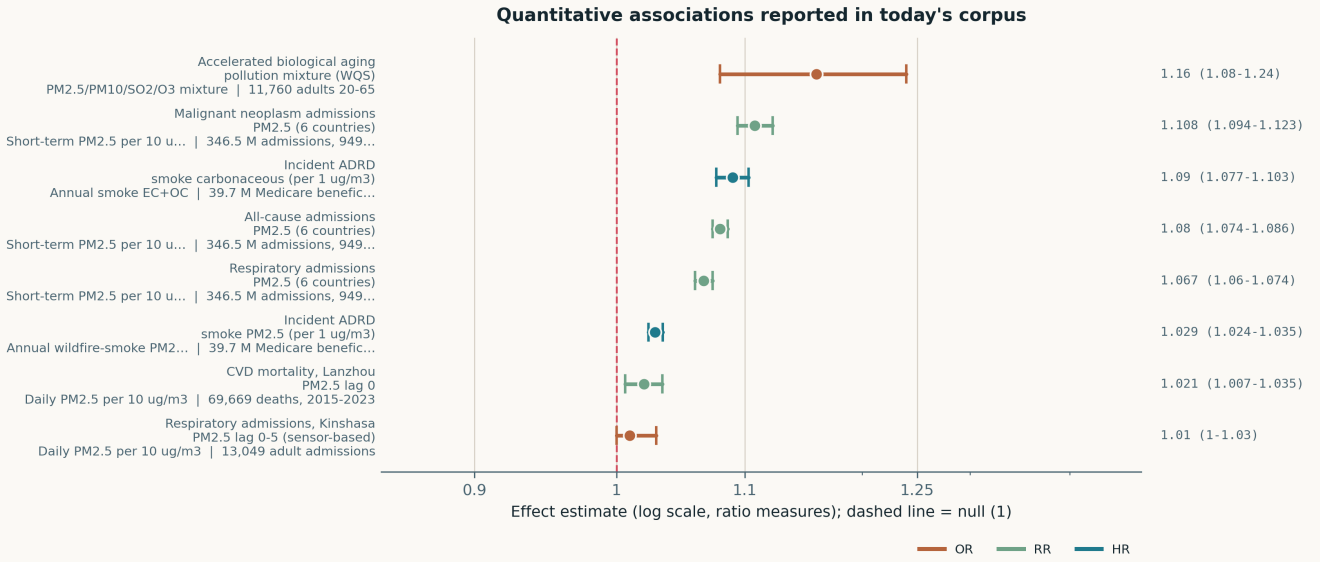


Fig. Eight ratio estimates with 95 % CIs. Scales differ and the panel is not a meta-analysis: three are per 10 μg m^{-3} short-term (multicountry admissions), two are per 1 μg m^{-3} long-term (Medicare smoke), one is a mixture-index odds ratio. The smoke-carbon HR is the largest per unit mass on the page; the Kinshasa OR is the only estimate built on low-cost sensor exposure, and its lower bound touches the null.

Paper-level digest

Ordered by cluster. Each entry gives the methodological core and the finding that matters, not an abstract paraphrase. Records marked *metadata only* carry no recoverable abstract and assert no finding.

Neuro / mental health — 2 records

Jin wildfire carbonaceous ADRD 2026 medRxiv (preprint)

Carbonaceous Components Underlying Wildfire Smoke $PM_{2.5}$-Associated Alzheimer's Disease and Related Dementias in Older US Adults

39.7 million fee-for-service Medicare beneficiaries, 2005-2019, 8.4 million incident ADRD cases; stratified Cox. Per 1 $\mu g\,m^{-3}$, smoke $PM_{2.5}$ HR 1.029 (1.024-1.035) for ADRD and 1.015 (1.005-1.026) for AD; the smoke carbonaceous fraction HR 1.090 (1.077-1.103) for ADRD. Per IQR the non-smoke association is larger (ADRD 1.048 vs 1.009) because smoke exposure contrast is narrow. Attributable: 11,142 ADRD cases/year. Limitation: preprint; annual smoke partitioning is model-derived and its error is not propagated, and claims-based incident dementia carries diagnostic-intensity confounding that tracks rurality and fire-prone regions. Relevance: high - argues for composition-resolved smoke metrics that optical LCS cannot supply without co-located carbon measurement.

doi: 10.64898/2026.10.08.26364863

Khan Pakistan neurovascular review 2026 Curr Neurovasc Res

Air Pollution and Neurovascular Health in Pakistan: An Emerging Public Health Crisis in South Asia - A Narrative Review

Narrative synthesis of mechanisms - systemic inflammation, oxidative stress, endothelial dysfunction, BBB disruption, microglial activation, prothrombotic change - linking particulate exposure to ischaemic and haemorrhagic stroke and small-vessel disease, with the observation that Pakistan has no longitudinal, biomarker or exposure-linked stroke surveillance data. Limitation: no search strategy, no new data, no quantitative synthesis; the gap statement is the contribution. Relevance: low-moderate - a call for registry-linked exposure that presupposes the monitoring network Pakistan does not have.

doi: 10.2174/0115672026460820260929113419

Cardiovascular & metabolic — 1 record

Chen Lanzhou $PM_{2.5}$ components CVD 2026 Int J Environ Health Res

Acute cardiovascular mortality risks from $PM_{2.5}$ and its chemical components in an industrial valley city: evidence from Lanzhou, China

69,669 CVD deaths, 2015-2023. $PM_{2.5}$ lag 0: excess risk 2.06% per 10 $\mu g\,m^{-3}$ (0.66-3.48%), larger in men and under-65s. qgcomp/WQS attribute risk to PAHs and Tl/Pb/Mn in traffic-dominated Chengguan, and to nitrate/chloride/sulfate and PAHs in industrial Xigu. Limitation: components are not measured but predicted by a random forest with r>0.70, so the mixture weights inherit the predictor set of that model - district contrasts may reflect which inputs drove the prediction rather than toxicity. Relevance: moderate - a valley inversion city where source-specific nodes would test exactly these district contrasts.

doi: 10.1080/09603123.2026.2745894

Respiratory & allergic — 3 records

Kinzunga Kinshasa $PM_{2.5}$ respiratory 2026 Research Square (preprint)

Acute Respiratory Impacts of Fine Particulate Matter ($PM_{2.5}$) in Kinshasa: Epidemiological Evidence and Pulmonological Implications

13,049 adult admissions at one hospital, 2018-2022; mean daily $PM_{2.5}$ 46-50 $\mu g\,m^{-3}$. Per 10 $\mu g\,m^{-3}$ over lag 0-5, adjusted OR 1.01 (1.00-1.03), strongest at 35-64 years and in Tshangu and Funa. Limitation: a single-hospital catchment against a city-wide exposure series, and an estimate whose lower bound sits on the null; the sensor and model fusion step is described but its validation against any reference in Kinshasa is not. Relevance: high - one of the first African health analyses whose exposure series is built on low-cost sensors, which makes the uncorrected sensor error the main threat to the inference.

doi: 10.21203/rs.3.rs-11301519/v1

Khan Gambia personal $PM_{2.5}$ pneumococcus 2026 medRxiv (preprint)

Longitudinal personal environmental exposure and Streptococcus pneumoniae colonisation in a cohort of children 5-14 years old in rural Gambia

134 children aged 5-14, 554 visits with paired personal exposure and culture. Geometric mean personal $PM_{2.5}$ 12.6 $\mu g\,m^{-3}$ (12.2-13.1), CO 0.16 ppm; colonisation 39.7%. Neither $PM_{2.5}$ nor CO was associated with colonisation after adjustment, including in within/between-child decomposition; between-child temperature was inversely associated (OR 0.74, 0.56-0.98). Limitation: personal $PM_{2.5}$ of 12.6 $\mu g\,m^{-3}$ in solid-fuel households is implausibly low for wood-smoke exposure and suggests the monitor or its wear compliance under-captured cooking peaks; a null under that exposure contrast is uninformative. Relevance: high - a clear case where personal-monitor validation, not the epidemiology, decides the result.

doi: 10.64898/2026.10.08.26364987

Trachalaki IPF $PM_{2.5}$ 2026 Ann Am Thorac Soc

Are Short-Term Increases in $PM_{2.5}$ Linked to Respiratory Hospitalizations in IPF?

METADATA ONLY. No abstract in PubMed, Europe PMC or Semantic Scholar at harvest; the title format suggests a research letter on short-term $PM_{2.5}$ and respiratory admissions in idiopathic pulmonary fibrosis. Recorded for coverage only - no result can be summarised, and none is inferred. Relevance: moderate if positive, because IPF is a small, identifiable cohort for which personal or home-sensor exposure is feasible.

doi: 10.1093/annalsats/aaoag317

Reproductive & developmental — 1 record

Liang CHD circulating proteins 2026 Nat Commun

A nested case-control study of air pollution and congenital heart defects via circulating proteins

Maternal periconception $PM_{2.5}$, PM_{10} and NO_2 were each associated with CHD, NO_2 most strongly; supervised sparse PLS identified pollutant-specific protein signatures in maternal and umbilical cord serum that tracked exposure and CHD, implicating FGF-4, VEGF-D, MSP and angiogenin. Limitation: the abstract gives no effect sizes or n; cord serum is sampled at delivery, months after cardiogenesis, so a cord-protein signature cannot be the mechanism of a first-trimester malformation, and supervised signature selection on the same outcome inflates association. Relevance: moderate - periconception exposure windows are short, which favours high-resolution residential estimates. Held from the 9 Oct issue.

doi: 10.1038/s41467-026-77623-9

Mechanistic toxicology — 3 records

Comis Turin oxidative potential 2026 Atmosphere

Oxidative Potential and Elemental Fingerprints of Urban and Suburban Particulate Matter in Turin

32 PM_{10} filters (urban traffic vs rural background, 2020-2021, min and max days of four months). Urban OP-AA 0.12-0.60 vs suburban 0.10-0.38 nmol AA/min/m3; Cu was the most active single metal. Total and phosphate-buffer bioaccessible metal fractions were compared. Limitation: filters were selected at monthly concentration extremes, so OP-mass relationships are inflated by design, and n=32 cannot support source attribution of OP. Relevance: low-moderate - OP remains unmeasurable by optical nodes; the Cu dominance points to brake-wear tracers as the practical proxy.

doi: 10.3390/atmos17101000

Yu alveolar organoid $PM_{2.5}$ 2026 Front Toxicol

Differential toxicity of water- and DMSO-extractable $PM_{2.5}$ fractions in human alveolar organoids

Human alveolar organoids exposed to extracted $PM_{2.5}$ showed dose- and time-dependent morphological damage, more severe for the DMSO-extractable (organic) fraction, with ROS rise, cell death and a sustained fall in AT2 cells plus a small AT1 increase suggesting AT2-to-AT1 transdifferentiation. Limitation: doses are not reported in deposited-mass terms, so relevance to ambient exposure is unknowable, and extraction solvents select chemistry rather than particles. Relevance: low - model development. Held from the 9 Oct issue.

doi: 10.3389/ftox.2026.1943932

Chen toluene soot toxicity 2026 Atmos Environ

Oxidative toxicity and cytotoxicity of toluene combustion soot particles mediated by their microstructure

METADATA ONLY. No abstract retrievable at harvest. From the title: soot generated from toluene combustion, with oxidative potential and cytotoxicity attributed to particle microstructure (likely graphitisation/nanostructure order). Recorded for coverage; no result is inferred. Relevance: low-moderate - if microstructure rather than mass sets toxicity, mass-based optical sensing misses a toxicity axis entirely.

doi: 10.1016/j.atmosenv.2026.122407

Exposure assessment & modelling — 6 records

Zhang Fusion-DeepRSM satellite $PM_{2.5}$ 2026 Environ Sci Technol

Fusion-DeepRSM for Satellite-Derived $PM_{2.5}$ Estimation and Precursor Sensitivity Prediction

Observation-constrained deep response-surface model fusing CTM runs, satellite products and ground $PM_{2.5}$. R2 0.872, RMSE 11.82 $\mu g\,m^{-3}$, and a 4.13 $\mu g\,m^{-3}$ reduction in rural overestimation, the known failure of station-trained ML. It also returns precursor-$PM_{2.5}$ response surfaces: for 2020-2023, meteorology offset part of emission-driven improvement (+1.2 vs -1.8 $\mu g\,m^{-3}$), and precursor change drove the Feb 2023 BTH rebound. Limitation: validation against stations that are themselves inputs; the rural bias correction is quantified on a CTM testbed, not against independent rural monitors - exactly where LCS would be needed. Relevance: high - a fusion architecture into which rural sensor nodes would slot as the missing constraint.

doi: 10.1021/acs.est.6c08137

Salem Riyadh $PM_{2.5}$ vegetation 2026 Environ Monit Assess

Spatiotemporal variation of $PM_{2.5}$ and its relationship with vegetation cover and meteorological factors during spring (2016-2022) in Riyadh, Saudi Arabia

16 stations, spring 2016-2022: seasonal mean $PM_{2.5}$ ranged from 94.6 (2018) to 55.0 $\mu g\,m^{-3}$ (2020), highest in southern and central Riyadh. Meteorological correlations were non-significant; the vegetation-$PM_{2.5}$ inverse association reached significance in one year only (rho=-0.539, p=0.031, n=16). Limitation: n=16 Spearman correlations per year cannot separate vegetation from land use, and dust events dominate spring variance; the one significant year out of seven is consistent with chance. Relevance: low-moderate - documents a dust-dominated regime where optical sensors' coarse-mode bias would govern any network.

doi: 10.1007/s10661-026-16057-8

Chen Chengdu WSON partitioning 2026 Atmos Chem Phys

Simultaneous measurements of gas- and aerosol-phase water-soluble organic nitrogen in the winter urban atmosphere of Chengdu, China

Winter urban campaign: particulate WSON 2.3+/-1.4 ugN/m3, twice gaseous, rising with $PM_{2.5}$; particle fraction 0.70+/-0.14. WSON partitions into organic phase under warm-dry and into aerosol liquid water under cold-humid conditions; amines are mainly gaseous but shift to particles as aminium sulfate/nitrate during episodes. ALWC dominated partitioning over T, RH and pH; $NH4NO_3$ was the main ALWC contributor. Limitation: one site, one season, and thermodynamic S-curve confirmation is model-on-model. Relevance: moderate - ALWC is precisely what inflates optical PM readings at high RH, so the nitrate-ALWC link is a calibration covariate as well as a chemistry result.

doi: 10.5194/acp-26-14329-2026

Li Guangzhou 3D vegetation 2026 Atmosphere

Scale-Dependent Effects of Three-Dimensional Urban Vegetation on Land Surface Temperature and Air Pollutants: Synergy and Trade-Off Zones in Guangzhou, China

Field LAI in 570 quadrats showed MODIS LAI is uninformative (r=0.15), so 3-D greenness was proxied by canopy height x tree cover. MGWR at 1-5 km: one SD of vegetation volume associated with 1.3 C lower LST, 0.30 $\mu g\,m^{-3}$ lower $PM_{2.5}$ and 1.3 $\mu g\,m^{-3}$ lower NO_2; O_3 null citywide with offsetting local responses. Limitation: $PM_{2.5}$ is a 1 km product whose own predictors include land cover, so the vegetation coefficient is partly the model reading its inputs back. Relevance: low-moderate - 0.3 $\mu g\,m^{-3}$ is below the error floor of any LCS network, which is itself the point.

doi: 10.3390/atmos17101001

Temirzhanova Pavlodar snow aerosol 2026 Atmosphere

Geochemical Characteristics of the Pollution of Snow Cover with Aerosols in an Industrial Center (Pavlodar, North-Eastern Kazakhstan)

55 snow samples across industrial, buffer, non-industrial and background zones. Pb exceeded background 2-100-fold, with Pb, Sr and V peaking in industrial and buffer zones; PC1 loaded on rare-earth elements. Limitation: snowpack integrates wet and dry deposition over an unknown accumulation period, so it is a deposition map, not an air concentration map, and cannot be converted to inhalation exposure. Relevance: low - useful as a cheap spatial prior for siting composition samplers where no network exists.

doi: 10.3390/atmos17100995

Wang China $PM_{2.5}$ decade 2026 Atmos Pollut Res

From rapid decline to slower improvement: Decadal nonlinear evolution and regional predictive associations of $PM_{2.5}$ in China (2015-2024)

METADATA ONLY. No abstract at harvest. Title indicates a nonlinear trend analysis of Chinese $PM_{2.5}$ showing deceleration of the post-2015 decline and regional predictors. Recorded for coverage. Relevance: low-moderate - slower improvement means smaller year-on-year signals, which raises the bar on sensor drift control for trend detection.

doi: 10.1016/j.apr.2026.103234

Sensing, forecasting & instrumentation — 4 records

Nobin Bangladesh GNN $PM_{2.5}$ 2026 PLoS One

Explainable spatiotemporal Graph Neural Networks: Multi-City $PM_{2.5}$ forecasting in Bangladesh with geographic graph topology validation

Graph-diffusion GRU on four years of daily multi-source data at 12 cities. Up to 3-day lead: MAE 8.85 $\mu g\,m^{-3}$, R2 0.666, beating non-graph baselines (FDR-corrected Diebold-Mariano p<0.001) and DCRNN when pooled across horizons. The useful result is negative for complexity: a fixed distance-adjacency graph beat adaptive, wind-informed and hybrid graphs, and occlusion-based influence tracked geographic adjacency (r=0.928). Limitation: 12 nodes is too sparse to discriminate graph topologies - transport structure that matters at 10-50 km scale cannot be resolved by stations hundreds of km apart. Relevance: high - a direct test of what graph priors a denser LCS network would need before learned adjacency pays off.

doi: 10.1371/journal.pone.0351928

Shai Nanozen RT vs gravimetric 2026 Ann Work Expo Health

Comparison of respirable dust concentrations between gravimetric weighing and real-time dust monitoring equipment in non-mining industries

20 paired task-based stationary samples at 11 industrial sites. No significant difference between gravimetric and raw real-time (Wilcoxon p=0.067) or density-adjusted (p=0.277); proportional bias grew with concentration but was absent below 1 mg/m3 (R2 0.06 and 0.02). Limitation: n=20, and the gravimetric reference is the instrument's own cassette downstream of the same inlet, so inlet losses and size-selection error are shared and invisible to the comparison; Wilcoxon non-significance at n=20 is not agreement. Relevance: moderate - the 1 mg/m3 cut is a usable operating limit, and the shared-inlet design is a pattern to avoid in LCS co-location.

doi: 10.1093/annweh/wxag086

Stutz Fairbanks inversion layers 2026 Atmos Chem Phys

Vertical distribution of pollution trapped by wintertime surface inversions in Fairbanks, Alaska, during the ALPACA experiment

Long-path DOAS profiles during ALPACA 2022 resolve polluted surface layers 20-40 m deep persisting for days, with SO_2 to 35 ppb, NO_2 60 ppb, HONO 2.5 ppb, HCHO 7 ppb and depleted ozone. PACT-1D, with mixing parameterised from surface temperature gradient and wind, reproduces the profiles and gives a 1-4 h residence time. Limitation: gas-phase only - particulate profiles, the health-relevant quantity in Fairbanks, are not retrieved - and one winter at one city. Relevance: high for siting - a 20-40 m layer means a node on a rooftop and one at breathing height sample different air masses under inversions.

doi: 10.5194/acp-26-14345-2026

He dust-storm PM_{10} deep learning 2026 Atmos Environ

Transport-aware spatiotemporal deep learning for accurate PM_{10} forecasts during dust storms

METADATA ONLY. No abstract at harvest. Title indicates a spatiotemporal deep-learning PM_{10} forecaster with explicit transport awareness, targeted at dust storms where station-only models lag the plume. Recorded for coverage; no metric is inferred. Relevance: moderate - upwind sensor placement is the obvious data complement to transport-aware models.

doi: 10.1016/j.atmosenv.2026.122409

Burden, policy & mitigation — 3 records

Yu multicountry $PM_{2.5}$ admissions 2026 Lancet Planet Health

Cause-specific hospital admissions associated with short-term $PM_{2.5}$ exposure: a multicountry time-series analysis

346.5 million admissions from 949 communities in six countries/territories, 2000-2019; quasi-Poisson time series pooled by meta-analysis. Per 10 $\mu g\,m^{-3}$, all-cause admissions RR 1.080 (1.074-1.086), with effects persisting to 21 days. Malignant neoplasms (1.108, 1.094-1.123), mental disorders (1.100) and nervous system disease (1.085) all exceed respiratory (1.067, 1.060-1.074). Attributable fraction against the WHO daily 15 $\mu g\,m^{-3}$ fell from 8.9% (2000-09) to 3.9% (2010-19). Limitation: a 21-day distributed lag on cancer admissions invites harvesting and scheduling artefacts - an elective oncology admission is not an acute PM event - and heterogeneity across countries is described as substantial but the pooled RR is reported without it. Relevance: high - the long-lag, non-respiratory signal argues for continuous rather than episodic sensing.

doi: 10.1016/j.lanplh.2026.101526

Bhattarai India crop burning inventories 2026 Atmos Environ

Air Quality Impacts of Post-monsoon Crop Residue Burning in Northern India Based on Regional and Global Emission Inventories

METADATA ONLY. No abstract at harvest. Title indicates a comparison of regional and global emission inventories for post-monsoon crop-residue burning and the resulting air-quality impact in northern India. Recorded for coverage. Relevance: moderate - inventory spread is the dominant uncertainty in Indo-Gangetic smoke attribution, and dense sensors are the observational constraint that could arbitrate between inventories.

doi: 10.1016/j.atmosenv.2026.122423

Sharma France net-zero benefits 2026 Environ Int

Health and economic benefits of air quality improvements in France through diverse net-zero transition scenarios by 2050

METADATA ONLY. No abstract at harvest. Title indicates a health-impact and economic assessment of air-quality co-benefits under several French net-zero pathways to 2050. Recorded for coverage; no estimate is inferred. Relevance: low - scenario work, but it fixes the concentration ranges a future network would need to resolve.

doi: 10.1016/j.envint.2026.110555

Other clinical endpoints — 1 record

Ma green space pollution aging 2026 Environ Health Perspect

Associations of Green Space and Air Pollution Mixture with Heterogeneous Biological Aging Patterns

11,760 adults aged 20-65 with repeated measures, accelerated longitudinal design; three BAA trajectories. Pollution mixture OR 1.16 (1.08-1.24) for accelerated-aging membership, driven mainly by O_3 and PM_{10}; NDVI OR 0.84 (0.75-0.95), 25% of which was mediated through lower pollution. Limitation: trajectories differ in level, not slope, so the design measures cross-sectional BAA differences dressed as trajectories; exposure contrast within one city is narrow and residential NDVI is strongly SES-patterned. Relevance: moderate - the narrow within-city contrast is the case for intra-urban sensor density. Held from the 9 Oct issue.

doi: 10.1021/ehp.6c00004

Corpus register

First author	Focus	Journal	Design	Metric	Region
Neuro / mental health					
Jin wildfire carbonaceous ADRD	Carbonaceous Components Underlying Wildfire Smoke $PM_{2.5}$-Associated Alzheimer's Disease and Related Dementias in Older US Adults	*medRxiv (preprint)*	Retrospective cohort	Annual smoke vs non-smoke $PM_{2.5}$ and smoke EC/OC (high-resolution model)	United States (Medicare, nationwide)
Khan Pakistan neurovascular review	Air Pollution and Neurovascular Health in Pakistan: An Emerging Public Health Crisis in South Asia - A Narrative Review	*Curr Neurovasc Res*	Narrative review	$PM_{2.5}$, PM_{10}, UFP (literature)	Pakistan
Cardiovascular & metabolic					
Chen Lanzhou $PM_{2.5}$ components CVD	Acute cardiovascular mortality risks from $PM_{2.5}$ and its chemical components in an industrial valley city: evidence from Lanzhou, China	*Int J Environ Health Res*	Time-series (GAM) with mixture models	Daily $PM_{2.5}$; random-forest-predicted components (PAHs, metals, ions)	China (Lanzhou)
Reproductive & developmental					
Liang CHD circulating proteins	A nested case-control study of air pollution and congenital heart defects via circulating proteins	*Nat Commun*	Nested case-control	Periconception $PM_{2.5}$, PM_{10}, NO2; maternal and cord serum proteomics	China
Respiratory & allergic					
Khan Gambia personal $PM_{2.5}$ pneumococcus	Longitudinal personal environmental exposure and Streptococcus pneumoniae colonisation in a cohort of children 5-14 years old in rural Gambia	*medRxiv (preprint)*	Prospective cohort	Personal 24-h $PM_{2.5}$ and CO monitors, repeated (8 visits)	Gambia (rural)
Kinzunga Kinshasa $PM_{2.5}$ respiratory	Acute Respiratory Impacts of Fine Particulate Matter ($PM_{2.5}$) in Kinshasa: Epidemiological Evidence and Pulmonological Implications	*Research Square (preprint)*	Time-stratified case-crossover	Daily $PM_{2.5}$ from PurpleAir and QuantAQ Modulair sensors plus bias-corrected GEOS-CF	Democratic Republic of the Congo (Kinshasa)
Trachalaki IPF $PM_{2.5}$	Are Short-Term Increases in $PM_{2.5}$ Linked to Respiratory Hospitalizations in IPF?	*Ann Am Thorac Soc*	Metadata only (no abstract)	Short-term $PM_{2.5}$	Not stated
Mechanistic toxicology					
Chen toluene soot toxicity	Oxidative toxicity and cytotoxicity of toluene combustion soot particles mediated by their microstructure	*Atmos Environ*	Metadata only (no abstract)	Laboratory toluene-flame soot	Laboratory
Comis Turin oxidative potential	Oxidative Potential and Elemental Fingerprints of Urban and Suburban Particulate Matter in Turin	*Atmosphere*	Chamber / laboratory	PM_{10} oxidative potential (ascorbic acid assay); total and bioaccessible metals	Italy (Turin)
Yu alveolar organoid $PM_{2.5}$	Differential toxicity of water- and DMSO-extractable $PM_{2.5}$ fractions in human alveolar organoids	*Front Toxicol*	In vitro exposure	Water- vs DMSO-extractable $PM_{2.5}$ fractions	Laboratory (human alveolar organoids)
Sensing, forecasting & instrumentation					
He dust-storm PM_{10} deep learning	Transport-aware spatiotemporal deep learning for accurate PM_{10} forecasts during dust storms	*Atmos Environ*	Metadata only (no abstract)	PM_{10} during dust storms	Not stated
Nobin Bangladesh GNN $PM_{2.5}$	Explainable spatiotemporal Graph Neural Networks: Multi-City $PM_{2.5}$ forecasting in Bangladesh with geographic graph topology validation	*PLoS One*	Machine-learning model	Daily $PM_{2.5}$, 12-city network, 1-3 day forecasts	Bangladesh (12 cities)
Shai Nanozen RT vs gravimetric	Comparison of respirable dust concentrations between gravimetric weighing and real-time dust monitoring equipment in non-mining industries	*Ann Work Expo Health*	Field measurement	Respirable dust: Nanozen DustCount 9000 optical vs its built-in filter cassette	South Africa (Gauteng)
Stutz Fairbanks inversion layers	Vertical distribution of pollution trapped by wintertime surface inversions in Fairbanks, Alaska, during the ALPACA experiment	*Atmos Chem Phys*	Field measurement	Vertical trace-gas profiles (LP-DOAS) with 1-D chemistry-transport model	United States (Fairbanks, Alaska)

First author	Focus	Journal	Design	Metric	Region
Exposure assessment & modelling					
Chen Chengdu WSON partitioning	Simultaneous measurements of gas- and aerosol-phase water-soluble organic nitrogen in the winter urban atmosphere of Chengdu, China	*Atmos Chem Phys*	Field measurement	Gas- and particle-phase WSON and amines; aerosol liquid water content	China (Chengdu)
Li Guangzhou 3D vegetation	Scale-Dependent Effects of Three-Dimensional Urban Vegetation on Land Surface Temperature and Air Pollutants: Synergy and Trade-Off Zones in Guangzhou, China	*Atmosphere*	Statistical model	1 km gridded $PM_{2.5}$, O3, NO2; layered vegetation volume	China (Guangzhou)
Salem Riyadh $PM_{2.5}$ vegetation	Spatiotemporal variation of $PM_{2.5}$ and its relationship with vegetation cover and meteorological factors during spring (2016-2022) in Riyadh, Saudi Arabia	*Environ Monit Assess*	Monitoring-record analysis	Spring $PM_{2.5}$ from 16 regulatory stations; fractional vegetation cover	Saudi Arabia (Riyadh)
Temirzhanova Pavlodar snow aerosol	Geochemical Characteristics of the Pollution of Snow Cover with Aerosols in an Industrial Center	*Atmosphere*	Field measurement	Elemental composition of snow-cover solid fraction (ICP-MS)	Kazakhstan (Pavlodar)
Wang China $PM_{2.5}$ decade	From rapid decline to slower improvement: Decadal nonlinear evolution and regional predictive associations of $PM_{2.5}$ in China	*Atmos Pollut Res*	Metadata only (no abstract)	Ambient $PM_{2.5}$ trends, 2015-2024	China (nationwide)
Zhang Fusion-DeepRSM satellite $PM_{2.5}$	Fusion-DeepRSM for Satellite-Derived $PM_{2.5}$ Estimation and Precursor Sensitivity Prediction	*Environ Sci Technol*	Machine-learning model	Gridded $PM_{2.5}$ from CTM + multisource satellite + ground fusion	China (nationwide)
Burden, policy & mitigation					
Bhattarai India crop burning inventories	Air Quality Impacts of Post-monsoon Crop Residue Burning in Northern India Based on Regional and Global Emission Inventories	*Atmos Environ*	Metadata only (no abstract)	Modelled $PM_{2.5}$ from crop-residue burning	India (northern)
Sharma France net-zero benefits	Health and economic benefits of air quality improvements in France through diverse net-zero transition scenarios by 2050	*Environ Int*	Metadata only (no abstract)	Scenario-modelled $PM_{2.5}$ and co-pollutants to 2050	France
Yu multicountry $PM_{2.5}$ admissions	Cause-specific hospital admissions associated with short-term $PM_{2.5}$ exposure: a multicountry time-series analysis	*Lancet Planet Health*	Multi-location time-series	Daily ambient $PM_{2.5}$ (community-level), lags to 21 days	Multi-country (Australia, Brazil, Canada, Chile, New Zealand, Taiwan)
Other clinical endpoints					
Ma green space pollution aging	Associations of Green Space and Air Pollution Mixture with Heterogeneous Biological Aging Patterns	*Environ Health Perspect*	Prospective cohort	$PM_{2.5}$, PM_{10}, SO2, O3 mixture (WQS); 500 m NDVI	China (single high-pollution city)

Provenance and method

Entry window **12 October 2026**, single day. Harvester legs: **PubMed** health **8**, sensing **0**; **Europe PMC 7** (4 AGRICOLA retro-index records dropped, 3 kept); **Crossref by ISSN 47**; **arXiv** no entry dated 12 October; **OpenAlex** HTTP 429. 58 unique, 0 already carried. **PubMed connector** (`[EDAT]` 12–14 October) returned 34 PMIDs, 8 dated 12 October, 3 appended — **61** candidates screened, **21 admitted**, **40 rejected** with logged reasons; plus **3 held records** from 9 October. Abstract retry (PubMed → Europe PMC → Semantic Scholar) recovered 6 of 13; **6 records carried on metadata alone**. **Consensus** calibration sweep: 10 + 10 hits, 0 new. **Scholar Gateway** still `ACCESS_DENIED`. Abstracts remain the copyright of their publishers.